

MAXWELL SHEPHERD

Bellevue, WA | +1 774 444 5952 | mshep0306@gmail.com | linkedin.com/in/maxwell-shepherd

EDUCATION

Johns Hopkins University

August 2024 - May 2026

M.S. Data Science with Honors

GPA: 4.00

- Specialization in Machine Learning & Artificial Intelligence
- Applied Generative AI, Computational Statistics, Big Data Engineering, Distributed Computing, Human-Computer Interaction

Selected Works:

- Contributing author, “A benchmark of expert-level academic questions to assess AI capabilities,” *Nature* (2026), via the HLE Contributors Consortium.
- “Frozen Vision Transformers for Dense Prediction on Small Datasets: A Case Study in Arrow Localization,” *arXiv:2604.16758* (2026).
- “Concurrent Streaming, Viewer Transfers, and Audience Loyalty in a Creator Ecosystem: A Minute-Level Longitudinal Study,” *arXiv:2603.23773* (2026).

Purdue University

August 2016 - May 2020

B.S. Industrial Engineering

GPA: 3.53

- Dean’s List & Semester Honors
- Operations Research, Optimization, Stochastic Modeling, Manufacturing Processes

EXPERIENCE

Boeing

August 2023 - Present

Industrial Engineer III / Engineering Data Scientist

- Built and deployed the organization’s first agentic AI / computer-use automation systems, delivering production agents to 6 teams that save an estimated 400 hours per week (and counting) in manual workflows.
- Delivered machine learning and statistical models for aircraft asset management on multi-terabyte datasets, influencing engineering resource allocation for 480 employees and driving schedule flow time reductions of 56%.
- Shipped self-serve analytics platform (Python, SQL, React) for enterprise-scale metrics, adopted by 1,100 employees and used daily by 260.
- Re-architected ETL pipelines processing 450M+ rows daily, cutting end-to-end latency from 24 hours to 15 minutes.
- Author of data synthesis and retrieval suite used by more than 1,000 employees 21,000 times per month, credited with a yearly cost savings of \$2.2M.

Boeing Designated Expert – Analytics

June 2025 - Present

- Recognized as enterprise-wide authority on business statistics, machine learning/GenAI, and analytics for an organization of 2,000+.
- Responsible for setting organization technical direction for future technologies and airplane programs.

Industrial Engineer (Intern, I, II)

May 2019 - August 2023

Disney – *Engineering Intern*

May 2018 - August 2018

PTC Inc. – *Engineering Intern*

May 2016 - August 2016

TECHNICAL SKILLS

AI Tools: Claude Code, Codex, Model Context Protocol (MCP)

Languages/Frameworks: Python (Flask, FastAPI, PyTorch, XGBoost), SQL, R, MATLAB, Minitab

Visualization: Tableau, Grafana

Ops: Docker, Airflow, Linux, InfluxDB, Git

Libraries: React (HTML/CSS/JS, Material-UI), OpenAI API, Claude API, Claude Agent SDK

Other: SolidWorks, CRDT (Conflict-free Replicated Data Types)